

Dashboard Metrics Display Fix

Date: November 9, 2025

Issue: Dashboard showing "Waiting for metrics" even though metrics file exists

Problem

The Streamlit dashboard was displaying:

```
Monitoring: logs/metrics
Waiting for: metrics_20251109.jsonl
```

Even though the metrics file existed and was being actively written to.

Root Cause

Path Mismatch: The dashboard was looking in the wrong directory.

- **Dashboard expected:** `/home/sujay/Programming/IDS/logs/metrics/`
- **Metrics actually in:**
`/home/sujay/Programming/IDS/dpdk_suricata_ml_pipeline/logs/metrics/`

The `logs/metrics/` directory exists but is empty. The actual metrics are in `dpdk_suricata_ml_pipeline/logs/metrics/`.

Solution

Fixed the `get_metrics_directory()` function in `dashboard.py`:

```
# OLD - WRONG:
def get_metrics_directory():
    script_dir = Path(__file__).parent
    return script_dir / 'logs' / 'metrics' # x Wrong path

# NEW - FIXED:
def get_metrics_directory():
    script_dir = Path(__file__).parent
    return script_dir / 'dpdk_suricata_ml_pipeline' / 'logs' / 'metrics'
# ✅ Correct path
```

Verification

```
# Check metrics file exists and is active
ls -lh
/home/sujay/Programming/IDS/dpdk_suricata_ml_pipeline/logs/metrics/metrics_20251109.jsonl
# Output: 64K file with 312+ lines

# Check file is being updated
tail -f
/home/sujay/Programming/IDS/dpdk_suricata_ml_pipeline/logs/metrics/metrics_20251109.jsonl
# Should see new lines appearing every ~15 seconds
```

Current Status

✅ **Metrics File:** Active and updating

- Location: `dpdk_suricata_ml_pipeline/logs/metrics/metrics_20251109.jsonl`
- Size: 64KB
- Lines: 312+
- Last update: Within last 30 seconds
- Component: `ensemble_consumer` (throughput metrics)

✅ **Dashboard:** Now pointing to correct directory

- Will auto-refresh every 5 seconds (default)
- Will show last 10 minutes of data (default)

What You Should See Now

After refreshing the dashboard (or waiting for auto-refresh):

PROF

1. **Overview Section:**

- Total Events Processed
- ML Predictions count
- Average Latency
- Total Errors

2. **Throughput Section:**

- Time series chart showing events/second
- Table showing `ensemble_consumer` throughput

3. **Other Sections** (if data available):

- Latency metrics
- ML predictions distribution

- System resources
- Errors/warnings

If Dashboard Still Shows "Waiting"

Possible reasons:

1. Dashboard hasn't auto-refreshed yet

- Click the "🔄 Refresh Now" button
- Or wait 5 seconds for auto-refresh

2. Time window too narrow

- Use sidebar slider to increase "Data lookback window" to 60 minutes
- This will show all metrics from the last hour

3. Need to restart dashboard

```
# Stop the dashboard (Ctrl+C in terminal)
# Restart it
streamlit run dashboard.py --server.port 8502
```

Monitoring in Real-Time

To verify metrics are flowing:

```
# Watch metrics file grow
watch -n 1 'ls -lh
/home/sujay/Programming/IDS/dpdk_suricata_ml_pipeline/logs/metrics/metri
cs_20251109.jsonl'

# See latest metrics in real-time
tail -f
/home/sujay/Programming/IDS/dpdk_suricata_ml_pipeline/logs/metrics/metri
cs_20251109.jsonl | jq .

# Count metrics by type
cat
/home/sujay/Programming/IDS/dpdk_suricata_ml_pipeline/logs/metrics/metri
cs_20251109.jsonl | \
jq -r .type | sort | uniq -c
```

Dashboard Configuration

Access via sidebar:

- **Auto-refresh interval:** 1-60 seconds (default: 5s)
- **Data lookback window:** 1-60 minutes (default: 10m)

Adjust these if you want:

- More frequent updates → decrease refresh interval
- More historical data → increase lookback window

Status:  Fixed - Dashboard now points to correct metrics directory